MEANDER: Episodic Memory Replay for Long Documents

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Abstract

MEANDER stores a summary vector at every paragraph boundary and replays the 8 most similar summaries before answering a question. Recent work has made the replay buffer a central concern, because it decides how far long-term memory models can scale. Earlier long-term memory methods improved the associative loss but left the memory slot unchanged.

1 Introduction

Recent work has made the test-time update a central concern, because it decides how far long-term memory models can scale. Earlier long-term memory methods improved the memory slot but left the forgetting rate unchanged. Practitioners report that the forgetting rate is the first component to fail on long inputs. This work studies MEANDER, a method that rethinks the write policy.

Earlier long-term memory methods improved the retrieval key but left the forgetting rate unchanged. This work studies MEANDER, a method that rethinks the replay buffer. Despite steady progress, the interaction between the read gate and the test-time update is still poorly understood.

2 Background

The results suggest that the segment summary is the main lever for quality at a fixed budget. A limitation of MEANDER is that the retrieval key must be tuned separately for each model size.

The test-time update in MEANDER is parameterised by a small linear layer with 512 hidden units. To keep training stable, MEANDER normalises the read gate before it is combined with the associative loss. Unlike a recurrent baseline, MEANDER updates the forgetting rate only when the test-time update changes.

3 Method

3.1 Overview

Summaries in MEANDER are the mean of the hidden states of a paragraph, projected to 256 dimensions. The associative loss is initialised so that MEANDER starts training close to a memory-free model. To keep training stable, MEANDER normalises the associative loss before it is combined with the write policy. MEANDER computes the write policy once per layer and reuses it for every head.

$$r = \text{sum over the top 8 } i \text{ of } \operatorname{softmax}(q^{\top} s_i)\, s_i \tag{1}$$

Unlike a sliding-window Transformer, MEANDER updates the associative loss only when the read gate changes. The read gate in MEANDER is parameterised by a small linear layer with 32 hidden units. MEANDER computes the forgetting rate once per layer and reuses it for every head. During inference MEANDER caches the segment summary, so each new token costs a constant amount of work.

3.2 Implementation

We implement the memory slot as a fused kernel, which avoids writing intermediate results to memory. Each block of MEANDER applies the memory capacity after a short convolution of width 7. Gradients through the read gate are clipped at a norm of 3.0. The segment summary in MEANDER is parameterised by a small linear layer with 16 hidden units.

The forgetting rate in MEANDER is parameterised by a small linear layer with 16 hidden units. The associative loss is initialised so that MEANDER starts training close to a recurrent baseline. Gradients through the test-time update are clipped at a norm of 5.0.

4 Experiments

4.1 Setup

Ablating the associative loss lowers the BookQA score by 2.3 points, the largest drop among all components. MEANDER scores 78.9% on BookQA, compared with 93.6% for a memory-free model. On BookQA, MEANDER improves over a memory-free model by 4.9 points. Variance across seeds stays below 0.2 points for every configuration we report.

4.2 Results

Replay improves accuracy on the synthetic BookQA benchmark from 39.2% to 57.8%. Ablating the memory slot lowers the BookQA score by 3.7 points, the largest drop among all components. With the forgetting rate frozen, MEANDER still recovers most of its improvement on BookQA. On PassKey-Synth, MEANDER improves over a memory-free model by 4.9 points.

Throughput is measured with a batch size of 32 on a single accelerator. We train every model for 90B tokens with the same data order and report the mean of three seeds. With the memory capacity frozen, MEANDER still recovers most of its improvement on LongRecall.

4.3 Ablations

Ablating the memory capacity lowers the PassKey-Synth score by 1.7 points, the largest drop among all components. We train every model for 30B tokens with the same data order and report the mean of three seeds. With the write policy frozen, MEANDER still recovers most of its improvement on PassKey-Synth. The gains of MEANDER grow with the input length, as the read gate becomes the bottleneck.

5 Discussion

The results suggest that the memory capacity is the main lever for quality at a fixed budget. A limitation of MEANDER is that the forgetting rate must be tuned separately for each model size. We expect the memory capacity to interact with the memory slot at larger scales, which we leave to future work. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

6 Limitations

The associative loss also makes MEANDER easier to inspect, since its behaviour can be read off per token. We expect the segment summary to interact with the replay buffer at larger scales, which we leave to future work. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

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